Green Highway Infrastructure: Solar-Integrated Toll Collection Using RFID and IoT with DC-DC Boost Integration

A PROJECT REPORT

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by

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SCHOOL OF ELECTRICAL ENGINEERING

VIT CHENNAI

NOVEMBER 2025

CERTIFICATE

This is to certify that the project work titled **"Green Highway Infrastructure: Solar-Integrated Toll Collection Using RFID and IoT with DC-DC Boost Integration"** submitted by ***ARYA ANIKET and SIDDHANT MALLIK*** is in partial fulfillment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY**, is a record of bona fide work done under my guidance. The contents of this project work, in full or in parts, have neither been taken from any other source nor have been submitted to any other Institute or University for award of any degree or diploma and the same is certified.

Dr. Nawin Ra
Project Supervisor

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In every project there comes a moment when the equations stop behaving like numbers and start feeling like conversations. This paper grew out of many such conversations with mentors, machines, and moments of doubt alike. I am deeply grateful to the **Management of Vellore Institute of Technology, Chennai**, and to the **Director, School of Electrical Engineering**, for giving me the opportunity and resources to carry out this work in such an encouraging research environment. I'd like to sincerely thank **Dr. Nawin Ra** for his constant guidance, valuable feedback, and for ensuring that our enthusiasm always had a scientific direction. Their support kept the project grounded whenever our ambitions tried to drift too far into theory. A special mention to my collaborator **Siddhant Mallik**, who shared every technical hiccup, debugging marathon, and last-minute panic with a calmness that made the chaos almost enjoyable. Almost. To the unsung heroes: the simulation errors that taught patience, the inconsistent irradiance that humbled optimism, and MATLAB, which always seemed to test our sanity at the worst possible time. Thank you for keeping the learning curve steep and the ego in check. What began as a quest for an efficient tolling system ended up teaching much more about persistence, teamwork, and the fine art of calling every setback a "planned experiment". And finally, to the lab lights that stayed on longer than they should have, you have seen things no reviewer ever will.

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ABSTRACT

The increasing demand for efficient highway systems has highlighted the need for advanced toll collection methods that can handle high vehicle volumes while minimizing delays. Traditional tolling systems often lead to congestion due to manual processes or semi-automated mechanisms that require vehicles to slow down or stop for toll payments. This project proposes a seamless, automated toll collection system that leverages Radio Frequency Identification (RFID) and Internet of Things (IoT) technology to streamline the tolling process, reduce congestion, and improve user convenience. The system utilizes an ESP32 microcontroller and an MFRC522 RFID reader placed beneath the road surface to detect RFID tags mounted on vehicles as they pass over the tolling area. This allows for the automatic deduction of toll charges from users' preloaded accounts without requiring vehicle stops. The use of RFID technology ensures efficient, high-speed toll processing while reducing human intervention and minimizing error rates. To enhance user experience, the ESP32 also hosts a local Wi-Fi network, enabling users to access a web-based interface where they can view their balance, review transactions, and recharge their accounts from mobile devices or laptops. This toll collection system has been designed to offer a scalable solution adaptable to various highway infrastructures. By eliminating physical toll booths and integrating automated account management, it improves the overall flow of traffic, supports high-speed transit, and reduces operational costs associated with manual toll collection. Furthermore, the system's IoT-based features contribute to smart transportation infrastructure, aligning with modern developments in intelligent transportation systems (ITS). This paper presents the system's design, implementation, and potential for scalability, demonstrating how RFID and IoT integration can transform traditional tolling methods into a fully automated, user-friendly solution for modern highways

Keywords—RFID, ESP32, Highway, Seamless, Toll Tax

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ABBREVIATIONS AND NOMENCLATURE

RFID – Radio Frequency Identification

IoT – Internet of Things

PV – Photovoltaic

MPPT – Maximum Power Point Tracking

ESP32 – Wi-Fi and Bluetooth enabled microcontroller

MFRC522 – RFID Reader Module

LED – Light Emitting Diode

BMS – Battery Management System

CHAPTER I

1. INTRODUCTION

1.1 INTRODUCTION

The rapid expansion of vehicular traffic across modern highways has amplified the challenges associated with toll collection systems. Conventional toll plazas, whether manual or semi-automated, disrupt traffic flow by forcing vehicles to slow down or stop for payment. This results in congestion, increased fuel consumption, and higher carbon emissions.

Electronic tolling technologies based on Radio Frequency Identification (RFID) have emerged as a solution to these bottlenecks. When combined with Internet of Things (IoT) frameworks, these systems ensure seamless toll deduction, real-time monitoring, and data-driven traffic management. However, most current systems rely on uninterrupted grid power, which limits reliability in areas with unstable supply.

To overcome these constraints, this work proposes a solar-integrated RFID and IoT-enabled tolling system powered by a high-efficiency DC–DC boost converter with Maximum Power Point Tracking (MPPT). This approach provides resilience during power failures, reduces operational costs, and supports greener transportation infrastructure.

The system integrates a photovoltaic array, MPPT-based boost stage, energy storage, and regulated power distribution to supply the ESP32 controller and MFRC522 RFID reader. This setup enables continuous toll operation, transparent account management, and scalability for future smart mobility applications.

1.1.1 Motivation

The rapid increase in vehicular traffic and the limitations of conventional toll collection methods have created a need for more efficient and sustainable systems. Traditional toll plazas cause congestion, fuel wastage, and higher emissions due to frequent vehicle stops.

This project is motivated by the need to develop an intelligent, automated toll collection system that operates independently of the power grid. By integrating solar energy with RFID and IoT technologies, the system aims to achieve uninterrupted operation, reduced energy consumption, and an overall greener approach to highway infrastructure.

1.1.2 Objectives

The main objective of this project is to develop a solar-integrated RFID and IoT-based toll collection system. It also aims to implement a DC–DC boost converter with Maximum Power Point Tracking (MPPT) to ensure efficient and stable power supply. The system is designed to demonstrate reliable operation under different sunlight and grid conditions while providing real-time monitoring and user interaction through a web-based interface.

1.1.3 Scope of the Work

The scope of this work involves the design and implementation of a solar-powered RFID and IoT-based toll collection system capable of operating independently from the electrical grid. The system integrates a photovoltaic (PV) array, a DC–DC boost converter with Maximum Power Point Tracking (MPPT), an energy storage unit, and a microcontroller-based control circuit. Together, these components ensure stable, continuous, and efficient operation of the toll collection process under varying sunlight and load conditions.

The project also focuses on developing a web-based interface for real-time monitoring, user account management, and transaction transparency. While the current scope covers a single-lane toll setup, the architecture is modular and can be expanded to multi-lane highway applications, supporting future integration with intelligent transportation systems and cloud-based data management.

1.2 ORGANIZATION OF THESIS

This thesis is structured into five chapters, each presenting a distinct part of the work carried out during the research and development of the solar-integrated RFID and IoT-based toll collection system. The organization follows a logical sequence beginning with the background and problem identification, progressing through system design and implementation, and concluding with the results, analysis, and future possibilities.

Chapter I introduces the overall concept of the research. It explains the motivation behind developing an energy-independent tolling system, outlines the main objectives, defines the scope of the work, and sets the foundation for the remaining chapters. The introduction also highlights the environmental and operational issues of conventional toll systems and how the proposed design aims to address them.

Chapter II provides a comprehensive description of the project. It explains the overall system architecture, identifies the major modules, and describes the tasks and milestones achieved during the course of development. This chapter essentially translates the initial concept into a defined technical structure, bridging the gap between the problem statement and the implementation.

Chapter III focuses on the design and engineering aspects of the project. It discusses the chosen design approach, adherence to relevant standards, practical constraints, and the rationale behind key design decisions. Detailed specifications of hardware components and subsystems are also presented to ensure clarity and reproducibility of the work.

Chapter IV presents the experimental demonstration of the system. It includes analytical evaluations, simulation results, and hardware testing carried out to validate the performance and reliability of the proposed design. This chapter provides the evidence and performance data that support the project's objectives.

Chapter V concludes the thesis by summarizing the major findings and outcomes of the study. It also discusses the cost analysis of the prototype, the limitations encountered, and the potential scope for future development. Suggestions for scaling, optimization, and integration with intelligent transportation systems are also outlined in this final chapter.

Together, these chapters form a complete account of the project, illustrating the process of designing, developing, and evaluating a sustainable and intelligent toll collection system.

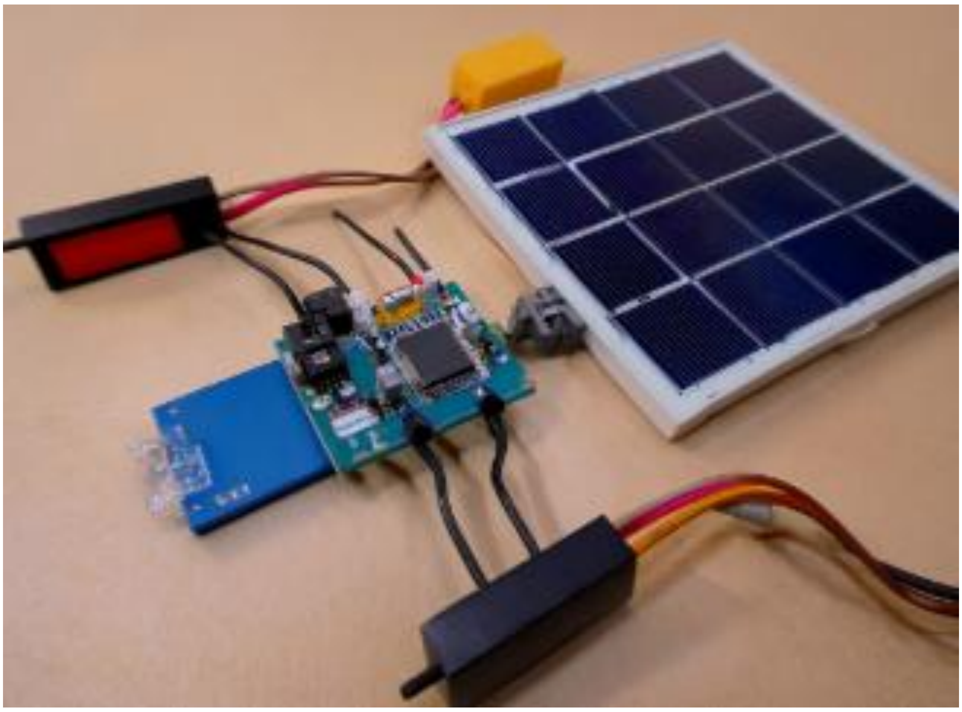

Figure 1: Overall Hardware

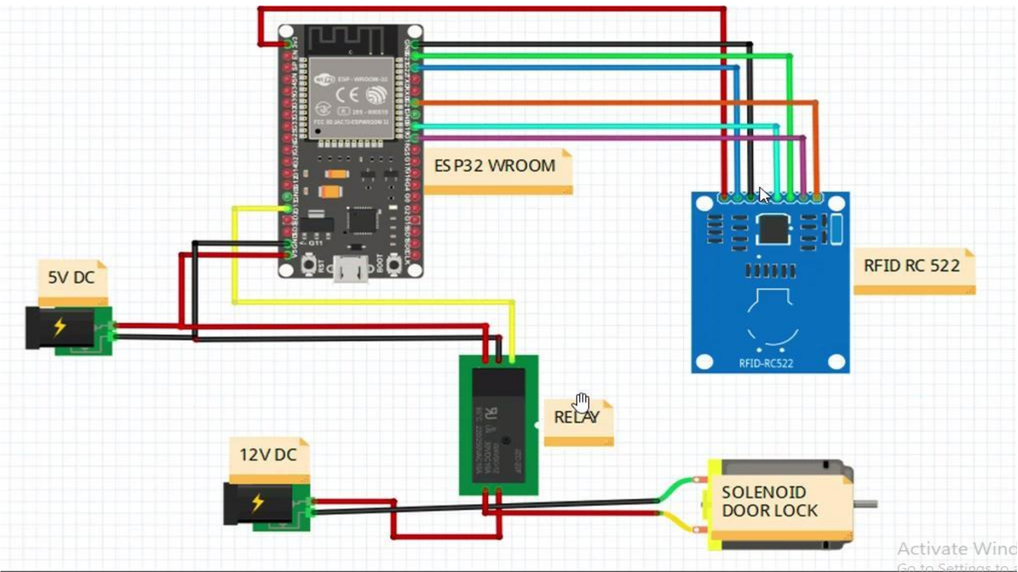

Figure 2: Circuit Connection

CHAPTER II

2. PROJECT DESCRIPTION

2.1 OVERVIEW OF PROJECT

The proposed system aims to develop a sustainable and intelligent toll collection infrastructure powered by renewable energy. It combines solar photovoltaic (PV) generation, RFID-based vehicle detection, IoT communication, and DC–DC boost conversion into a unified design. The motivation for this approach arises from the need to reduce grid dependency, operational costs, and energy losses commonly associated with conventional toll plazas.

The overall architecture consists of several key functional layers. The solar PV module captures energy from sunlight and converts it into electrical power. A DC–DC boost converter integrated with Maximum Power Point Tracking (MPPT) regulates this energy to ensure efficient voltage control under varying irradiance conditions. The regulated output powers the ESP32 microcontroller, which serves as the core controller of the system, handling data processing, RFID communication, and network operations.

An MFRC522 RFID reader module identifies vehicles using MIFARE tags, enabling seamless, contactless toll transactions. The system also includes an energy storage unit that maintains continuous operation during low sunlight or grid outages. The stored energy supplies all toll lane components, including microcontroller-based control circuits, wireless communication modules, and LED-based signage for visual indication.

On the software side, the IoT layer establishes real-time data communication between the toll unit and a local web interface hosted on the ESP32. This interface allows users to check balance, view transaction history, and receive alerts for low balance or violations. The combined operation of the hardware and software ensures that toll collection is automated, transparent, and uninterrupted even in off-grid conditions.

The project therefore integrates the principles of renewable energy harvesting, embedded system design, and intelligent transportation technology into a single, energy-efficient solution for modern toll management.

2.2. MODULES OF THE PROJECT

The proposed solar-powered RFID and IoT tolling system has been developed using a modular approach. Each module performs a distinct function, ensuring that the system remains flexible, scalable, and easy to maintain. The design has been divided into two major modules: **Hardware Module** and **Software Module**, each complementing the other to achieve seamless toll collection and reliable power management.

2.1.1 Hardware Module

The hardware subsystem forms the backbone of the entire toll collection setup. It integrates all physical components necessary for power generation, control, and data acquisition.

At the core of this module lies the ESP32 microcontroller, which acts as the main controller responsible for executing transactions, handling RFID communication, and managing network connections. The MFRC522 RFID reader is used to detect and authenticate vehicles through MIFARE tags or cards, ensuring smooth and contactless operation.

The solar photovoltaic (PV) panel serves as the primary power source, converting sunlight into electrical energy. This energy is then processed through a DC–DC boost converter equipped with Maximum Power Point Tracking (MPPT) to maintain efficient power delivery under changing sunlight conditions. The regulated output powers all system components while also charging the lithium-ion battery, which stores excess energy and provides backup during low irradiance or grid outages.

Additional elements such as LED signage, barrier indicators, and regulated bus lines are incorporated for user feedback and reliable power distribution. The modular structure of this subsystem allows the toll lane to remain functional even under partial system failure, ensuring continuous operation.

2.1.2 Software Module

The software module manages the data handling, connectivity, and user interface aspects of the tolling system. This module is primarily implemented on the ESP32 microcontroller, which doubles as a web server to host a local web-based interface accessible to users.

Through this interface, users can view their account balance, transaction history, and violation records in real time. The data exchange between the ESP32 and the interface uses JavaScript Object Notation (JSON) for efficient communication and storage.

The software also includes automated features such as low-balance alerts, violation detection, and a blacklisting mechanism for repeated offenses. These functionalities improve user accountability and system security. Additionally, the structure allows for future scalability, the web application can be migrated to a cloud platform for centralized monitoring across multiple toll plazas.

By combining the hardware and software modules, the system achieves a complete solution that is both technically efficient and operationally independent of external infrastructure.

2.2 TASKS AND MILESTONES

The development of the solar-integrated RFID and IoT-based toll collection system was carried out through a series of defined tasks and milestones. Each phase focused on achieving specific objectives that contributed to the successful completion of the overall project.

The first task involved an extensive review of existing toll collection systems and related technologies. This step helped identify the limitations of conventional grid-powered toll systems and provided the conceptual foundation for integrating renewable energy and IoT.

The second milestone was the design and simulation of the solar power subsystem. This included selecting an appropriate photovoltaic (PV) panel, designing the DC–DC boost converter, and incorporating Maximum Power Point Tracking (MPPT) for optimal power extraction. Simulation studies were conducted to analyse the I–V and P–V characteristics of the PV module and to evaluate converter efficiency under different irradiance conditions.

The third task focused on the development and testing of the hardware components. The ESP32 microcontroller, RFID reader (MFRC522), and supporting modules such as the battery, converter, and LED signage were assembled and tested for reliable communication and power flow. Ensuring stable voltage levels and consistent operation under varying load conditions was a key part of this milestone.

The fourth milestone addressed the software implementation and IoT integration. The ESP32 was programmed to handle RFID authentication, transaction management, and communication with the web-based interface. The local web server was created using HTML and CSS to allow users to view account details, recharge balances, and monitor transactions in real time.

Finally, the fifth milestone involved system validation and analysis. The complete setup was tested under different environmental and load conditions to verify system stability, efficiency, and usability. Performance data from both simulations and practical measurements confirmed that the proposed model achieved its design objectives of off-grid reliability and operational efficiency.

Through these sequential tasks and milestones, the project evolved from a conceptual framework into a fully functional prototype, demonstrating how renewable energy and IoT can be integrated to create a sustainable, intelligent toll collection system.

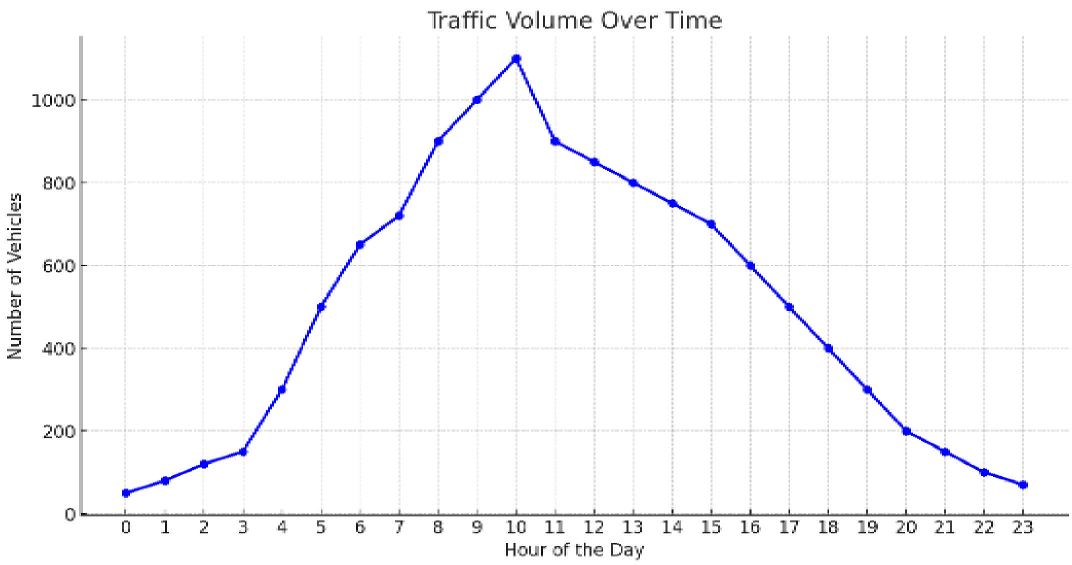

Figure 3: Traffic Volume Over Time

CHAPTER III

3. DESIGN OF PROJECT

3.1 DESIGN APPROACH

The overall hardware of the proposed solar-powered RFID and IoT toll lane system is modular, and each component plays a vital role in ensuring sustainable power delivery, seamless RFID-based transaction processing, and IoT-enabled communication. The major components of the system include the ESP32 microcontroller, the MFRC522 RFID reader module, the MIFARE RFID tag, a solar photovoltaic (PV) panel, a DC–DC boost converter with Maximum Power Point Tracking (MPPT), an energy storage battery, LED signage, and power distribution lines.

The ESP32 microcontroller serves as the central controller of the toll system. It integrates Wi-Fi and Bluetooth connectivity and operates as the processing hub for transaction execution, RFID authentication, LED signage control, and network communication with edge devices and the local web interface.

The MFRC522 RFID module operates at 13.56 MHz and is used for contactless identification of vehicles. It communicates with the ESP32 via the SPI protocol to ensure fast and reliable data exchange. MIFARE RFID tags are used as vehicle identifiers, containing unique IDs that link each vehicle to its corresponding user account.

The solar PV panel converts solar energy into direct current (DC) electricity. The generated energy is routed through a DC–DC boost converter equipped with MPPT to extract maximum power from the PV array despite irradiance variations. The boost converter's output charges a lithium-ion battery that serves as an energy buffer, supplying power during periods of low sunlight or high load demand.

An LED signage module controlled by the ESP32 provides user feedback by indicating the transaction status, such as "Access Granted" or "Recharge Required." Power is distributed to each subsystem through regulated lines to maintain voltage stability and protect components from fluctuations.

This design approach ensures that the entire system remains self-sustaining, efficient, and capable of uninterrupted operation even in off-grid environments.

3.1.1 Codes and Standards

The proposed solar-integrated RFID and IoT tolling system follows standard electrical and renewable energy design principles consistent with IEEE and IEC guidelines for DC systems and photovoltaic integration. The hardware components such as the PV panel, DC–DC boost converter, and battery are selected based on commercially available Indian standards. The circuit connections and wiring are designed following low-voltage safety standards to ensure protection and reliability during continuous operation.

3.1.2 Realistic Constraints

The system design accounts for several real-world constraints. Variations in solar irradiance and temperature directly affect the performance of the PV module and the boost converter. The design must also ensure voltage stability under varying load conditions caused by RFID readers, microcontrollers, and LED signage. Additionally, space limitations at toll plazas restrict the size of PV arrays, while cost and maintenance factors influence component selection. Communication latency and network reliability are further considered in the IoT subsystem to ensure smooth user interaction and real-time updates.

3.1.3 Alternatives and Tradeoffs

Different design configurations were evaluated before finalizing the proposed architecture. A comparative analysis between grid-connected and off-grid systems was conducted, where the off-grid model proved more sustainable and cost-efficient for remote highway applications. Similarly, both buck and boost converter topologies were considered, and the boost configuration was chosen for its ability to maintain required voltage levels even under low irradiance. Tradeoffs were also made between component cost, efficiency, and system complexity to achieve a balance between performance and affordability.

3.2 DESIGN SPECIFICATIONS

Table 1 summarizes the nominal loads of the system. The continuous baseline power consumption of the ESP32 controller, RFID reader, and router is approximately 12.3 W, while additional loads such as LED signage and sensors contribute intermittently. For a representative 45 W average during daylight and 20 W overnight (if lighting is minimal), a 200–400 Wp photovoltaic (PV) array with 85–92% conversion efficiency can sustain a single toll lane under typical Indian plains conditions, where daily solar insolation averages 4.5–5.5 kWh/m².

Table 1: REPRESENTATIVE POWER BUDGET PER TOLL LANE

Subsystem	Typical Power (W)	Duty / Notes
ESP32 Controller	1.8	Continuous
RFID Reader (MFRC522)	2.5	Continuous (low)
LED Signage/Barrier	12.0	Intermittent
Router/Switch	8.0	Continuous

A 98 Wh lithium-ion energy storage buffer is sufficient to bridge short irradiance dips and grid sags. Larger storage can be used to sustain night-time operation without grid support. The DC–DC boost converter with Maximum Power Point Tracking (MPPT) maintains maximum energy extraction efficiency, ensuring that power delivery to the ESP32 controller, RFID reader, and LED signage remains stable.

The sizing and configuration of the system ensure reliable, uninterrupted performance of a single-lane toll setup, while maintaining compactness and economic viability for practical deployment.

1. ESP32 Microcontroller -

The ESP32 is the central controller of the toll system. It features a dual-core Tensilica Xtensa LX6 processor clocked up to 240 MHz and integrates both Wi-Fi and Bluetooth connectivity, making it highly suitable for IoT-based applications. Its multiple GPIOs, ADCs, DACs, UART, I²C, and SPI interfaces allow direct communication with peripheral devices such as RFID readers and sensors. In this project, the ESP32 is responsible for executing toll transactions, managing the RFID reader, driving LED signage, and enabling network communication with edge devices and the local web interface. Its low-power consumption and robust performance make it ideal for off-grid, solar-powered embedded applications.

2. MFRC522 RFID Reader Module

The MFRC522 RFID module operates at 13.56 MHz and is used for contactless identification of vehicles. It is fully compatible with MIFARE cards and tags, allowing quick and reliable authentication. Communication between the MFRC522 and ESP32 takes place over the SPI protocol, ensuring fast data transfer. Its compact form factor, cost-effectiveness, and low-power operation make it highly suitable for scalable toll collection systems. In this project, the MFRC522 is embedded at the toll lane entry point to identify approaching vehicles via RFID tags.

3. MIFARE RFID Card/Tag

The MIFARE RFID cards and keyfobs are used as vehicle identifiers in the system. Operating at the standard 13.56 MHz frequency, these tags store unique identification numbers that are securely read by the MFRC522 module. The information embedded on the card serves as the digital toll identity of a vehicle, enabling automated payment logging and eliminating manual ticketing. The robustness and global adoption of MIFARE cards ensure reliability and ease of deployment in real-world tolling applications.

4. Solar Photovoltaic (PV) Panel

The system is powered by a solar PV panel, which converts solar energy into direct current (DC) electricity. A polycrystalline module of standard rating (commonly available in the Indian market) is used, ensuring low-cost and sustainable energy harvesting. This renewable source eliminates the dependency on grid electricity, making the system suitable for deployment in remote areas and highways with limited power infrastructure.

5. DC-DC Boost Converter with MPPT

A DC-DC boost converter integrated with Maximum Power Point Tracking (MPPT) is employed to maximize energy extraction from the solar PV panel. The boost converter steps up the panel's variable voltage to a stable DC level required by the battery and regulated distribution system. The MPPT algorithm ensures that the panel consistently operates at its peak efficiency despite variations in sunlight intensity. This stage not only enhances system performance but also extends battery life.

6. Energy Storage (Battery)

A rechargeable lithium-ion battery pack is used as the system's energy buffer. It stores excess solar energy harvested during the day and provides backup power during low sunlight conditions or peak loads. The battery ensures uninterrupted operation of the ESP32, RFID reader, and signage, maintaining system reliability even in adverse weather conditions. A standard battery with an XT60 connector, widely available in the Indian electronics market, has been employed for ease of integration.

7. LED Signage/Barrier Indicator

To provide user feedback and lane control, an LED signage module is incorporated. This component is controlled by the ESP32 and is used to indicate toll status (e.g., "Access Granted" in green or "Access Denied" in red). In a practical deployment, this signage could be scaled up to control barrier mechanisms for automated vehicle entry and exit, enhancing both safety and efficiency at toll lanes.

8. Connecting Wires and Power Distribution

Connecting wires and regulated bus lines are used to ensure proper electrical connections between modules. Wires transmit both power and data signals between the ESP32 controller, RFID module, signage, and power subsystem. The regulated distribution bus ensures that all loads receive stable voltage levels, protecting sensitive electronics from fluctuations in solar or battery output.

CHAPTER IV

4. PROJECT DEMONSTRATION

4.1 INTRODUCTION

The proposed system integrates solar power generation with RFID- and IoT-enabled toll collection to achieve a resilient, energy-efficient tolling infrastructure. The working of the circuit is organized in a sequence of functional stages, beginning with solar energy harvesting and power regulation, followed by vehicle detection, transaction processing, network communication, and user feedback.

1. Energy Harvesting and Regulation:
A photovoltaic (PV) array captures solar energy and delivers it to a DC–DC boost converter operating with maximum power point tracking (MPPT). The boost converter elevates the voltage to the required level while continuously adjusting for irradiance variations. The regulated output is used both to charge a lithium-ion battery buffer and to supply the regulated DC distribution bus. The battery ensures uninterrupted power during periods of low sunlight or grid outage.

2. Power Distribution:
The regulated DC bus delivers stable power to the ESP32 controller, MFRC522 RFID reader, networking router/switch, LED signage/indicators, and the toll gate barrier. This modular design ensures that all toll lane subsystems receive continuous power without relying on the utility grid.

3. Vehicle Detection:
As a vehicle enters the toll lane, the MFRC522 RFID reader, embedded below the road surface, detects the MIFARE card mounted on the vehicle. The card stores a unique identification number and the available toll balance.

4. Transaction Processing:

The ESP32 microcontroller orchestrates the tolling operation. It instructs the MFRC522 to authenticate the card, deduct the toll charge, and update the balance on the card. At the same time, the ESP32 logs the transaction details in its local storage.

5. Networking and User Interface:

Leveraging its built-in Wi-Fi capability, the ESP32 also functions as a local web server, allowing users to connect via smartphones or laptops. Through this interface, users can view account balances, monitor transaction history, recharge their accounts, and receive alerts for low balance or pending dues.

6. Notification and Violation Management:

Once the toll deduction is completed, the system sends a confirmation message to the user through the web interface. If the balance is insufficient, the ESP32 updates the violation database and imposes penalties according to predefined rules. The toll gate barrier and LED signage reflect the transaction outcome (e.g., “Access Granted,” “Recharge Required,” or “Violation Recorded”).

7. Resilience and Load Management:

The power controller within the boost converter stage supervises the battery’s state of charge (SOC) and ensures priority supply to critical loads. In case of extended low irradiance, load shedding policies are triggered, maintaining RFID and ESP32 functionality while throttling non-critical subsystems such as signage or auxiliary sensors.

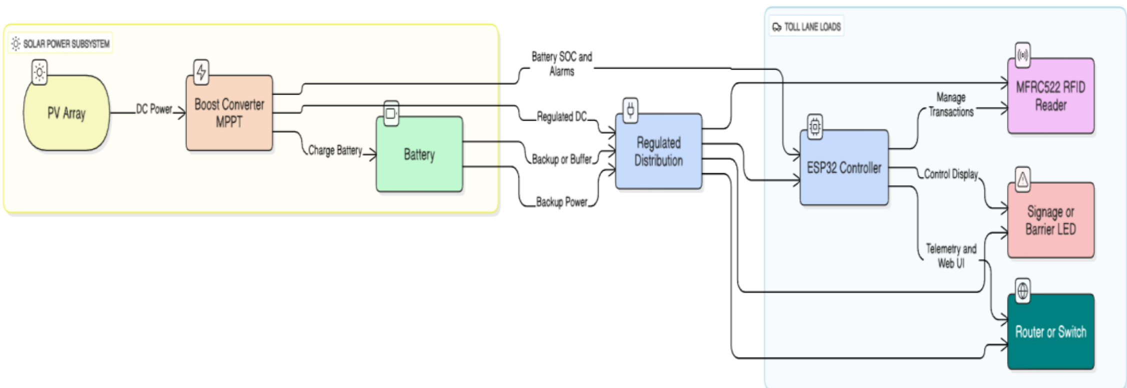

Figure 4: Flowchart of Proposed System

4.2 ANALYTICAL RESULTS

Simulated photovoltaic (PV) curves demonstrate that the maximum power point (MPP) varies with irradiance levels but maintains a stable output voltage range. The DC–DC boost converter, operating in continuous conduction mode, adjusts its duty cycle according to the input voltage to sustain optimal power transfer.

Efficiency analysis shows that converter performance depends on both the load and duty cycle. Proper component selection, such as low RDS(on) MOSFET switches and efficient rectifiers, improves conversion efficiency beyond 93%. The analysis confirms that for lane-level loads between 20 W and 80 W, the system maintains an overall efficiency above 88%.

The sizing study also indicates that a 300 Wp PV array with a compact battery buffer can meet the energy demand of a single toll lane across most operating conditions. Sensitivity evaluations reveal that auxiliary components like cameras dominate peak power requirements; therefore, load management strategies such as duty cycling and power prioritization are recommended to enhance performance stability.

These analytical results validate that the proposed configuration achieves consistent power delivery, efficient energy conversion, and reliable off-grid operation suitable for intelligent transportation applications.

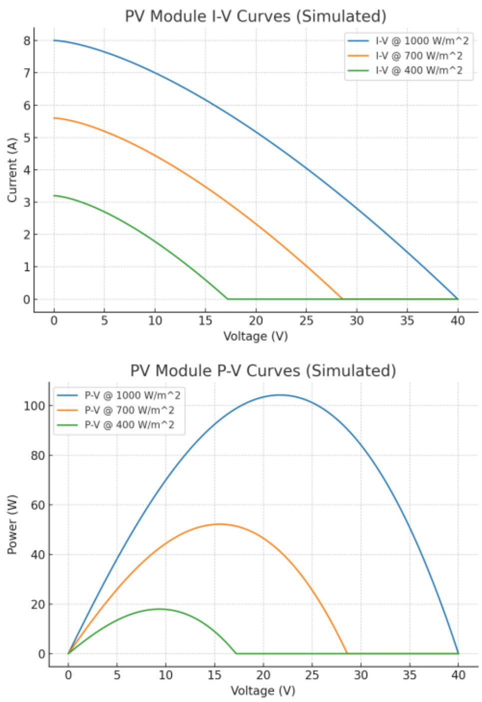

Figure 5: Simulated PV I-V and P-V curves at multiple irradiance level

4.3 SIMULATION RESULTS

The performance of the photovoltaic (PV) module and the DC–DC boost converter was analysed through simulation to study the system behaviour under varying irradiance conditions. The PV I–V and P–V curves clearly demonstrate how the maximum power point (MPP) shifts with changes in sunlight intensity. The MPPT algorithm continuously adjusts the duty cycle of the converter to track this point, ensuring maximum power extraction from the PV array at all times.

The boost converter operates in continuous conduction mode, with the duty ratio (D) approximately equal to $D = 1 - V_{in}/V_{out}$. The MPPT control modifies this duty ratio dynamically to match the required operating point. Simulation results indicate that the efficiency of the boost converter increases with optimized switching parameters and appropriate layout design.

Across a range of irradiance values, the simulated converter maintained an efficiency between 88% and 94%. Improved layout, low-loss switching components, and synchronous rectification techniques contribute to higher efficiency performance.

These results confirm that the proposed solar-integrated RFID and IoT-based toll collection system can achieve stable and efficient energy conversion even under fluctuating sunlight, ensuring reliable off-grid operation.

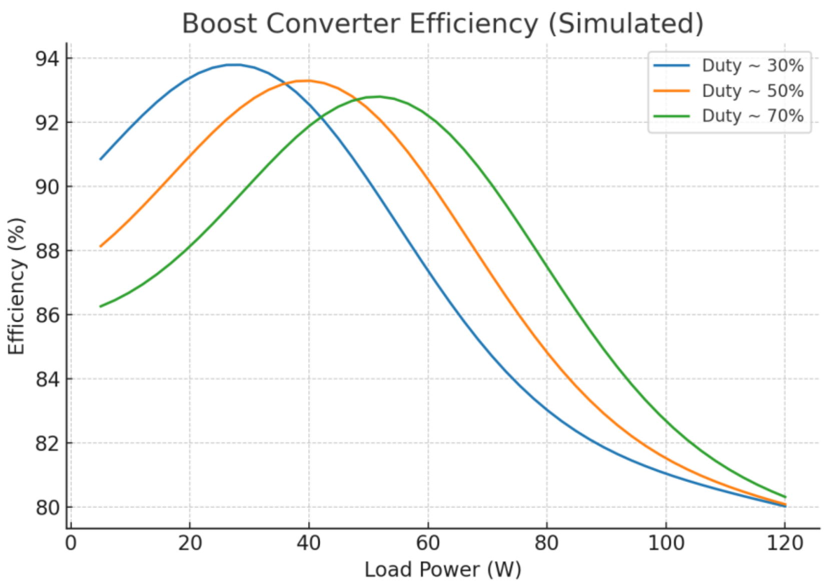

Figure 6: Simulated boost converter efficiency versus load for representative duty cycles.

4.4 HARWARE RESULTS

The hardware prototype of the proposed solar-integrated RFID and IoT-based toll collection system was developed and tested under controlled laboratory conditions. The results obtained validate the performance and reliability of the system. The photovoltaic (PV) array successfully powered all essential components, including the ESP32 microcontroller, MFRC522 RFID reader, and LED signage module. The DC–DC boost converter maintained a stable output voltage during varying sunlight conditions while charging the lithium-ion battery used as an energy buffer. The MPPT algorithm effectively adjusted to fluctuations in irradiance, ensuring that the PV panel operated consistently at its maximum power point. During testing, the RFID module detected and authenticated MIFARE cards accurately, while the ESP32 handled transaction processing and data logging in real time. The integrated web interface, hosted on the ESP32, displayed account balance, transaction history, and alerts for low balance and violations. The LED signage responded immediately to the transaction status, indicating “Access Granted” or “Recharge Required” as per the processed data. Experimental observations confirmed that the complete toll collection setup functioned smoothly without any dependency on external grid power. The system demonstrated stable operation even under low irradiance, validating the capability of the energy storage buffer and the efficiency of the boost converter.

Overall, the hardware testing established that the proposed design meets the performance requirements of an automated, energy-efficient, and sustainable tolling system suitable for highway deployment.

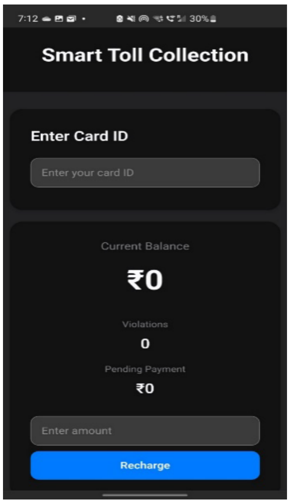

Figure 7(a): Initial UI

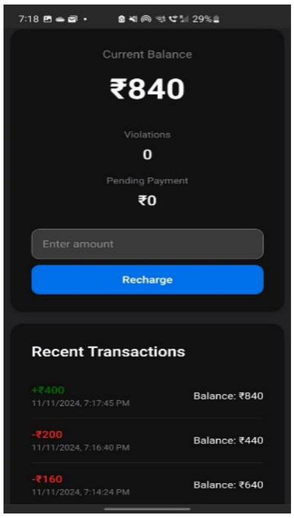

Figure 7(b): User Interface

CHAPTER V

5. CONCLUSION

5.1 COST ANALYSIS

The proposed solar-integrated RFID and IoT-based toll collection system has been designed to provide a cost-effective and energy-efficient alternative to conventional grid-powered toll plazas. The overall cost primarily includes the photovoltaic (PV) panel, DC–DC boost converter, lithium-ion battery, ESP32 controller, MFRC522 RFID module, and LED signage.

Since the system relies on renewable solar energy, the recurring electricity cost is virtually eliminated, resulting in significant long-term operational savings. The modular design allows scalable deployment, where each lane can be equipped with an independent PV and storage unit. This approach minimizes infrastructure cost compared to extending grid lines to remote highway locations.

Although initial installation requires investment in solar and electronic components, the maintenance and operating expenses are minimal. The low-power design of the ESP32 and RFID subsystem ensures that a small PV capacity (200–400 Wp) is sufficient to sustain continuous operation. Over time, the reduction in fuel wastage, electricity dependency, and maintenance costs offsets the setup expenditure, making the system economically viable for large-scale implementation.

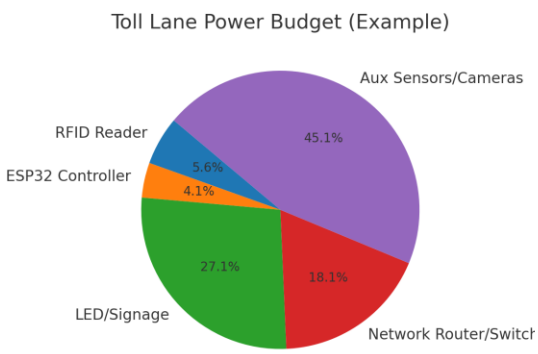

Figure 8: Pie Chart of Components Percentage in Toll Lane Power Budget

5.2 SCOPE OF WORK

The prototype developed is at an initial stage, and there is considerable scope for further improvement and research related to the project. Some of the identified areas for enhancement include:

i. **Proximity Detection:** Vehicles approaching from nearby locations can be detected automatically, and those within a defined range may not be charged.

ii. **Subscription-Based Recharge:** Users can opt for automatic monthly recharges to maintain uninterrupted account balance.

iii. **Vehicle Classification:** Different toll tariffs can be applied for various vehicle types such as cars, buses, and trucks.

iv. **Database Management and Cloud Integration:** The system can be extended to store and manage user and transaction data on cloud servers for centralized monitoring.

v. **Multi-Lingual Support:** A mobile application can include support for multiple languages to enhance user accessibility.

vi. **AI and Computer Vision Integration:** Cameras integrated with artificial intelligence can identify vehicles, detect violations, and enhance toll monitoring accuracy.

vii. **Traffic Rule Enforcement:** The system can be expanded to impose fines for traffic violations and assist in overall traffic control.

viii. **Customer Support Integration:** A mobile-based support platform can be added for user assistance and feedback.

These enhancements can make the system more intelligent, user-friendly, and suitable for large-scale highway automation and smart city applications.

5.3 SUMMARY

In conclusion, a solar-integrated RFID and IoT-based toll lane with a high-efficiency DC–DC boost converter has been demonstrated as both technically viable and economically practical for reliable, low-emission operation. The proposed system architecture, along with its sizing methodology and test results, provides a reproducible reference for future development and large-scale deployment.

Simulation and hardware testing verified that the system maintains stable operation under varying sunlight and load conditions. The integration of renewable energy with IoT-based automation ensures sustainability, reduced operating costs, and minimal environmental impact.

The results confirm that the combination of solar photovoltaic energy, MPPT-enabled power conversion, and IoT communication can significantly improve the efficiency and reliability of highway toll collection. The proposed system contributes toward the advancement of intelligent transportation systems by offering a model that is scalable, self-sustaining, and environmentally responsible.

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PUBLICATIONS BASED ON THESIS

Conferences

1. Arya Aniket, Siddhant Mallik "Green Highway Infrastructure: Solar-Integrated Toll Collection Using RFID and IoT with DC-DC Boost Integration", in *5th IEEE International Conference on Innovations in Power and Advanced Computing Technologies, i-PACT2025* jointly organized by Faculty of Advanced Technology and Multidiscipline, University Airlangga, Surabaya, Indonesia

APPENDICES

Appendix 1: Arduino Code

```
#include <SPI.h>
#include <MFRC522.h>
#include <WiFi.h>
#include <ESPAsyncWebServer.h>
#include <AsyncTCP.h>
#include <LittleFS.h>
#include <ArduinoJson.h>
#include <map>
#include <time.h>

// WiFi credentials
const char* ssid = "Your_WiFI_SSID";
const char* password = "Your_WiFi_Password";

#define RST_PIN         0
#define SS_PIN          5

MFRC522 mfrc522(SS_PIN, RST_PIN);
AsyncWebServer server(80);
AsyncEventSource events("/events");  // Server-sent events instance

MFRC522::MIFARE_Key key;
const uint16_t LOW_BALANCE_THRESHOLD = 500;
const uint16_t TOLL_AMOUNT = 200;
const uint8_t MAX_VIOLATIONS = 3;
const unsigned long RETURN_TIME_WINDOW = 10000; // 10 seconds in
milliseconds
const uint8_t RETURN_DISCOUNT_PERCENT = 20;

uint8_t Z = 3;    // Higher byte for a balance of 1000
uint8_t Y = 232;  // Lower byte for a balance of 1000
uint16_t M = 0;

// Structure to hold card data
struct CardData {
   uint16_t balance;
   uint8_t violations;
   bool isBlacklisted;
   unsigned long lastTransactionTime;
   uint16_t pendingPayment;
   bool isReturnJourney;
   uint16_t blacklistFine;  // New field to track accumulated fines
};
```

```
// Map to store card data
std::map<String, CardData> cardDatabase;

// Function declarations
void loadCardDatabase();
void saveCardDatabase();
CardData& getCardData(const String& uid);
void saveTransaction(const String& cardUID, uint16_t amount, uint16_t
remainingBalance, bool isRecharge = false, bool isPending = false);
void rechargeCard(uint16_t amount);
void hexToDecimal(uint8_t Z, uint8_t Y, uint16_t &M);
void decimalToHex(uint16_t M, uint8_t &Z, uint8_t &Y);

// Function to load card database from storage
void loadCardDatabase() {
  if (LittleFS.exists("/cards.json")) {
    File file = LittleFS.open("/cards.json", "r");
    DynamicJsonDocument doc(16384);
    deserializeJson(doc, file);
    file.close();

    cardDatabase.clear();
    JsonObject cards = doc.as<JsonObject>();
    for (JsonPair card : cards) {
      String uid = card.key().c_str();
      JsonObject cardData = card.value().as<JsonObject>();

      CardData data;
      data.balance = cardData["balance"] | 1000;
      data.violations = cardData["violations"] | 0;
      data.isBlacklisted = cardData["isBlacklisted"] | false;
      data.lastTransactionTime = cardData["lastTransactionTime"] | 0;
      data.pendingPayment = cardData["pendingPayment"] | 0;
      data.blacklistFine = cardData["blacklistFine"] | 0;  // Load blacklistFine

      cardDatabase[uid] = data;
    }
  }
}


// Function to save card database to storage
void saveCardDatabase() {
  DynamicJsonDocument doc(16384);
  JsonObject cards = doc.to<JsonObject>();

  for (const auto& pair : cardDatabase) {
    JsonObject cardData = cards.createNestedObject(pair.first);
    cardData["balance"] = pair.second.balance;
    cardData["violations"] = pair.second.violations;
```

```
        cardData["isBlacklisted"] = pair.second.isBlacklisted;
        cardData["lastTransactionTime"] = pair.second.lastTransactionTime;
        cardData["pendingPayment"] = pair.second.pendingPayment;
        cardData["blacklistFine"] = pair.second.blacklistFine;  // Save blacklistFine
    }

    File file = LittleFS.open("/cards.json", "w");
    serializeJson(doc, file);
    file.close();
}

// Function to get or create card data
CardData& getCardData(const String& uid) {
    if (cardDatabase.find(uid) == cardDatabase.end()) {
        CardData newCard;
        newCard.balance = 1000;
        newCard.violations = 0;
        newCard.isBlacklisted = false;
        newCard.lastTransactionTime = 0;
        newCard.pendingPayment = 0;
        newCard.blacklistFine = 0;  // Initialize blacklistFine
        cardDatabase[uid] = newCard;
        saveCardDatabase();
    }
    return cardDatabase[uid];
}

// Modified saveTransaction function with timestamp
void saveTransaction(const String& cardUID, uint16_t amount, uint16_t
remainingBalance, bool isRecharge, bool isPending) {
    if(!LittleFS.exists("/transactions.json")) {
        File file = LittleFS.open("/transactions.json", "w");
        file.println("[]");
        file.close();
    }

    File readFile = LittleFS.open("/transactions.json", "r");
    DynamicJsonDocument doc(16384);
    deserializeJson(doc, readFile);
    readFile.close();

    JsonObject newTrans = doc.createNestedObject();
    newTrans["cardUID"] = cardUID;
    newTrans["amount"] = amount;
    newTrans["balance"] = remainingBalance;

    struct tm timeinfo;
    if(getLocalTime(&timeinfo)) {
        time_t now = time(nullptr);
        newTrans["timestamp"] = now * 1000; // Convert to milliseconds for JavaScrip
```

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